Pitch — Aufgabenblatt 3

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Agenda

What I did to solve the tasks

Goals and Scope

Mapping / Planning

Obstacle Detection

- Active Map Update
- Replanning

Conclusion

Goals and Scope

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Scope

What goals did the task sheet demand

Block A	Block B
Arena mapping	LiDAR obstacle integration
Path planning	Active map updating
	Collision-avoiding navigation

Block A: Mapping / Planning

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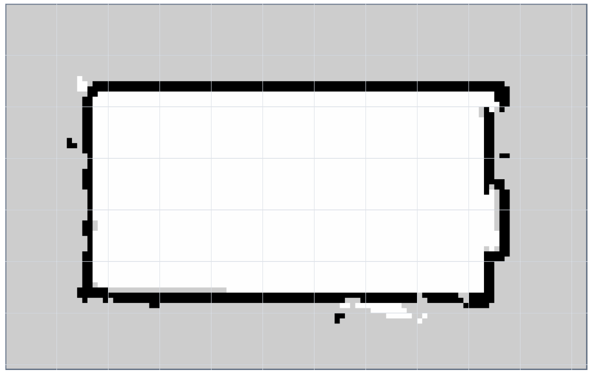
First problem
A map is not yet a planning map



maps/aufgabe03/arena_1p898x3p9_auto.yaml

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This is the stationary map created from SLAM.

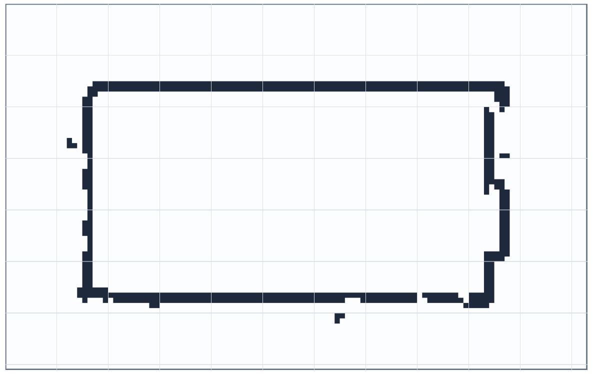
The SLAM result is only a saved occupancy map.

Occupancy mask

Map is converted into a trinity occupancy grid

Cell types:

- Free
- Occupied
- Unknown



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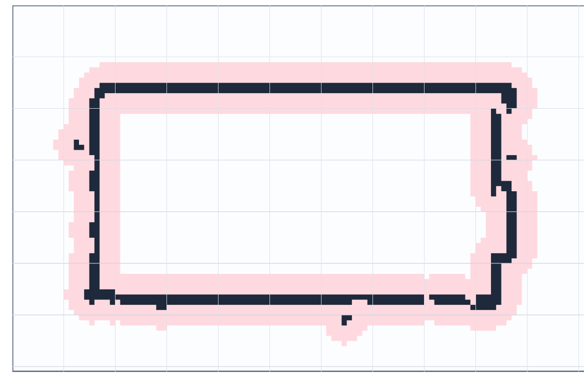
For planning I first had to convert it into a trinary grid, classify free/occupied/unknown cells.

Inflated blocked mask

Inflate obstacles and walls by robots clearance radius

Resolution $= 0.05 \, \frac{m}{\text{cell}}$



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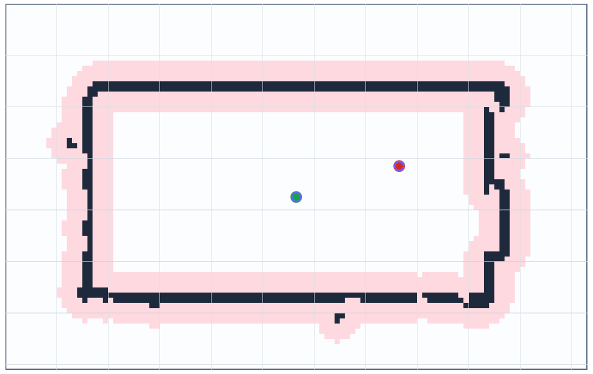
Then I had to inflate walls by the robot clearance.

This immediately changed which paths were actually valid.

Start and goal snapping

Planner snaps start and goal to traversable cells



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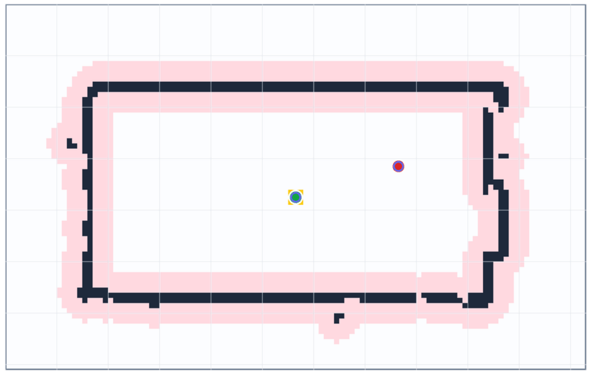
Offline A* search expansion

Initial route is planned on the saved SLAM grid before execution

- 8-neighbor motion
- Euclidean heuristic
- Inflated obstacle mask



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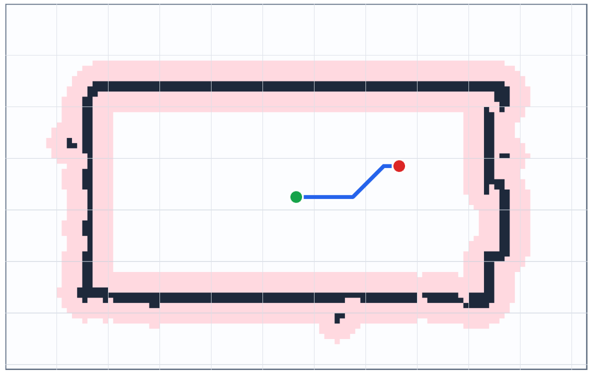
For the initial route, I used an offline A* planner on the saved occupancy map.

This gave me full control over the occupancy conversion, wall inflation, start/goal snapping, and waypoint simplification.

Later, the same planning idea is reused for replanning on the temporary run-local obstacle map.

Final dense path

Choosing path found by A*



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A* gives a dense grid path, but the robot cannot execute every grid cell directly.

Simplified waypoints

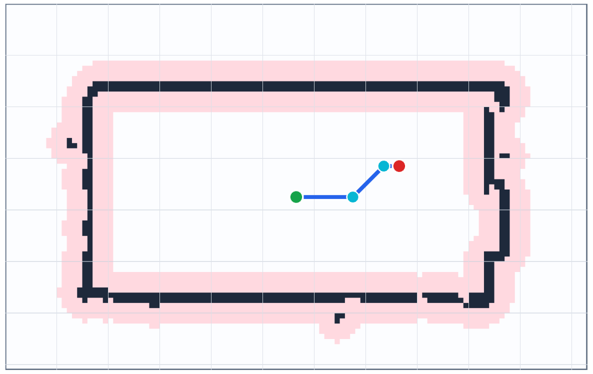
Grid path is simplified into executable waypoints

Dense path: 21 cells

Simplified waypoints: 4

Path length: ~1.124 m

linear_speed_mps = 0.03



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I had to simplify the path into waypoints while keeping the route collision-free.

The planned path only became usable after tuning the follower.

Faster controller settings caused failures/timeouts; slower motion was reliable enough to complete the route.

Block B: Obstacle detection

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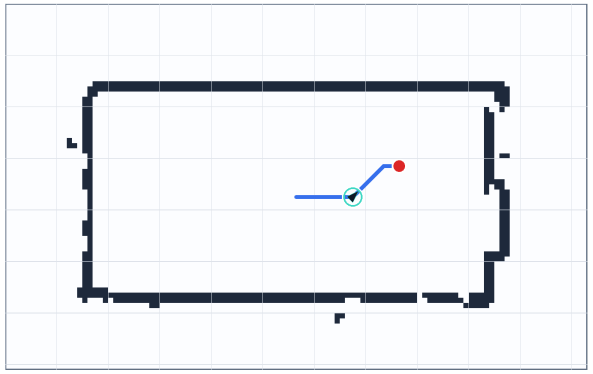
Robot follows planned route

Same map and path using the waypoints



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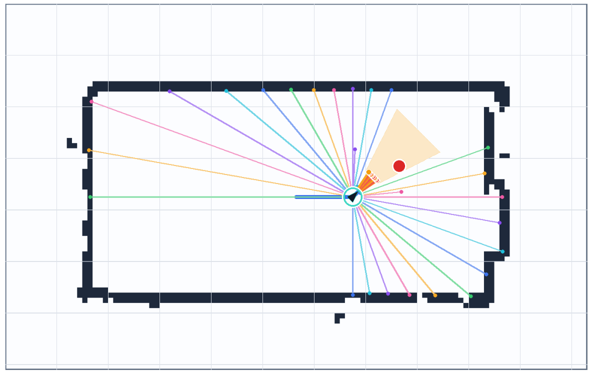
Raw LiDAR scan points

Robot receives raw /scan returns during motion



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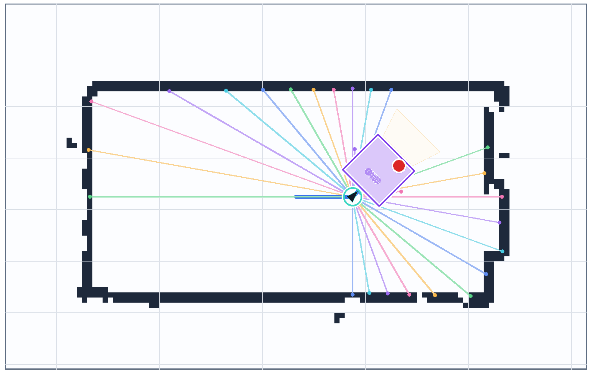
Filter points through forward ROI

Only relevant points in front of the robot are considered



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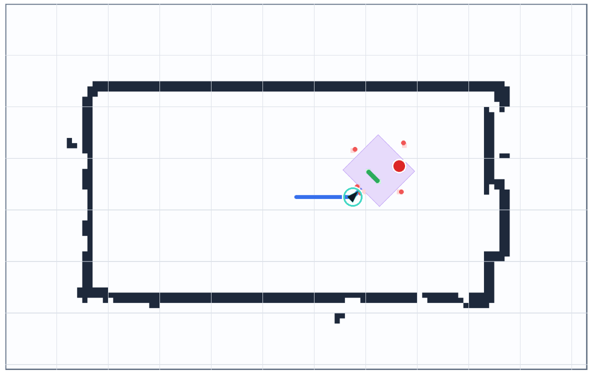
Accepted vs rejected cells

Rejected points near static walls or outside bounds



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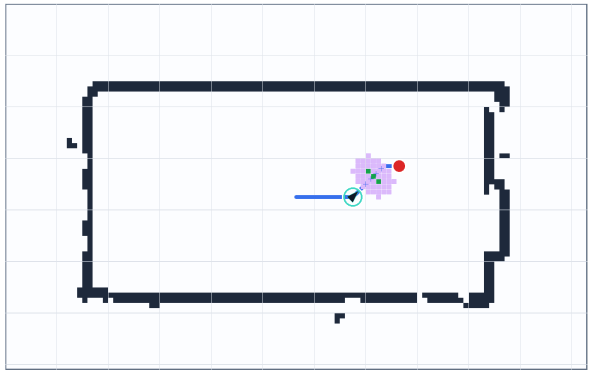
Local obstacle overlay

Temporary run-local overlay is added



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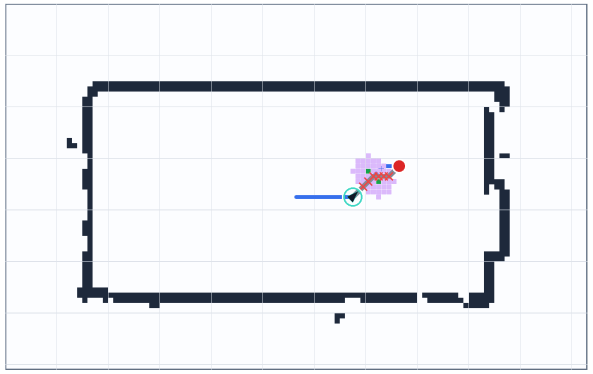
Old path blocked

Inflate the obstacle for clearance



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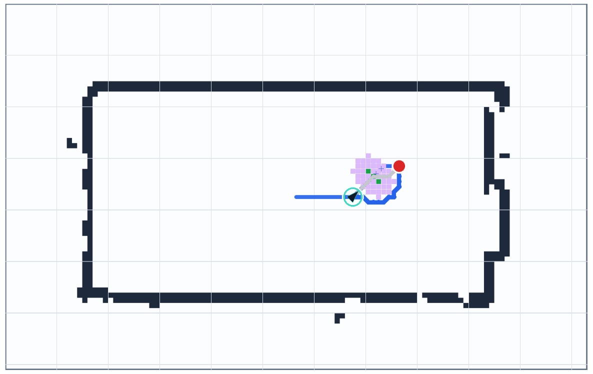
Replanned path

Robot stops, updates temporary map, replans, and resumes



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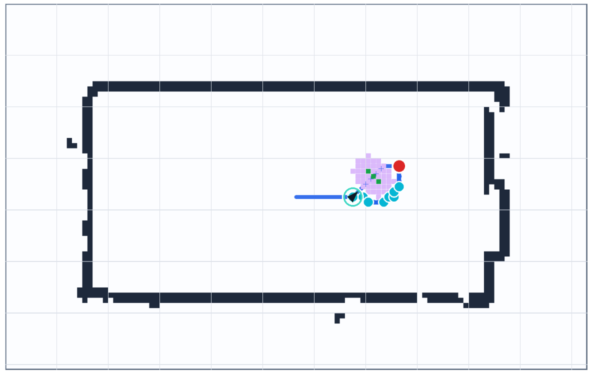
Validation

Select final route with new waypoints



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Conclusion

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